AIR POLLUTION TREND ANALYSIS IN DELHI USING AQI DATA

Name : B. Sony

Domain : Data Science

Shadow-Fox Virtual Internship Program

Introduction

Air pollution is a major environmental challenge affecting millions of people worldwide. Delhi frequently experiences poor air quality due to increasing urbanization, industrial emissions, vehicular pollution, and unfavorable weather conditions. Monitoring pollutant concentrations helps in understanding air quality patterns and identifying potential health risks.

This project focuses on analyzing air pollution trends in Delhi using pollutant concentration data. The study examines pollutant behavior over time, identifies peak pollution periods, and investigates relationships between different pollutants to gain meaningful environmental insights.

Objectives

- To study air pollution trends in Delhi using AQI-related data.
- To identify pollutants with the highest concentration levels.
- To determine the most polluted periods during the observation window.
- To analyze hourly variations in pollution levels.
- To investigate relationships between major air pollutants.

Dataset Overview

The dataset contains hourly air quality observations collected from Delhi during January 2023.

```
Shape: (561, 9)

Columns:
Index(['date', 'co', 'no', 'no2', 'o3', 'so2', 'pm2_5', 'pm10', 'nh3'], dtype='object')
```

	date	co	no	no2	o3	so2	pm2_5	pm10	nh3
0	2023-01-01 00:00:00	1655.58	1.66	39.41	5.90	17.88	169.29	194.64	5.83
1	2023-01-01 01:00:00	1869.20	6.82	42.16	1.99	22.17	182.84	211.08	7.66
2	2023-01-01 02:00:00	2510.07	27.72	43.87	0.02	30.04	220.25	260.68	11.40
3	2023-01-01 03:00:00	3150.94	55.43	44.55	0.85	35.76	252.90	304.12	13.55
4	2023-01-01 04:00:00	3471.37	68.84	45.24	5.45	39.10	266.36	322.80	14.19

Dataset Summary

- Number of Records: 561

- Number of Features: 9

- Observation Type: Hourly Air Quality Measurements

Variables Included

- Date and Time

- Carbon Monoxide (CO)

- Nitric Oxide (NO)

- Nitrogen Dioxide (NO_2)

- Ozone (O_3)

- Sulfur Dioxide (SO_2)

- PM2.5

- PM10

- Ammonia (NH_3)

Data Preparation

The dataset was prepared for analysis using Python.

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Preprocessing Steps

- Imported dataset using Pandas.
- Verified dataset dimensions and structure.
- Checked for missing values.
- Checked for duplicate records.
- Converted date information into datetime format.
- Extracted hourly and daily information from timestamps.
- Generated AQI indicators for pollution analysis.

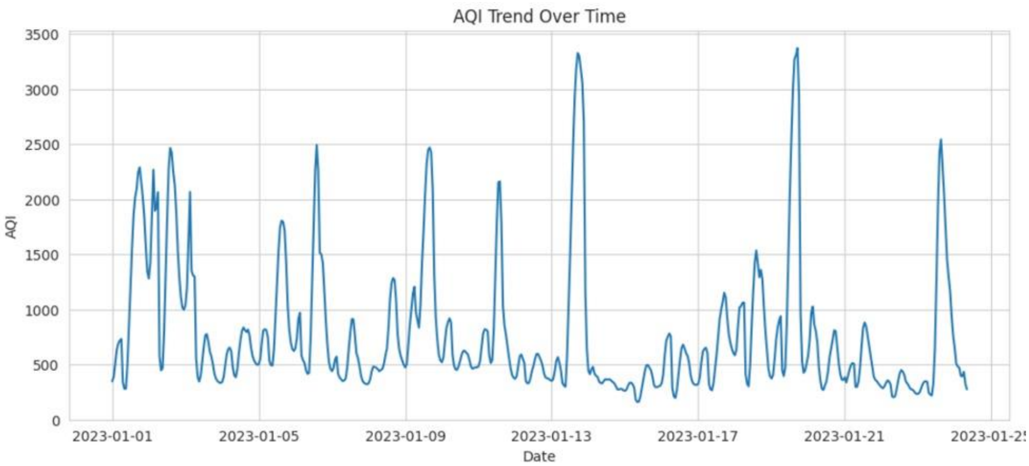
Data Quality

The dataset was found to be complete with no missing values and no duplicate observations.

Analysis and Visualizations

AQI Trend Analysis

The AQI trend visualization highlights fluctuations in pollution levels throughout the observation period.



Findings

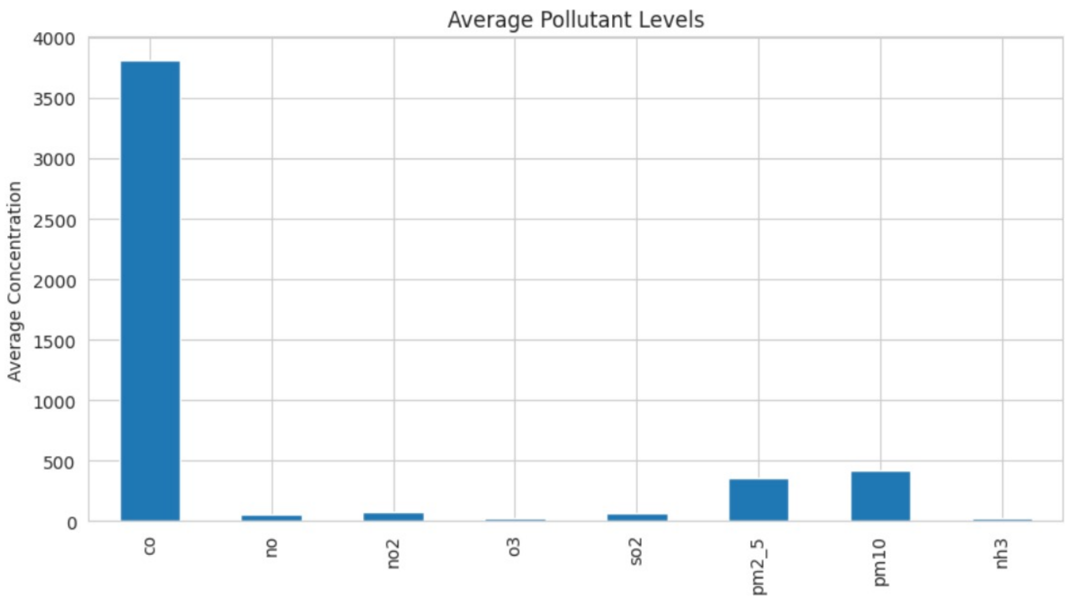
- Pollution levels varied considerably across different days.

- Several high-pollution episodes were identified.

- AQI peaks occurred during specific periods in January.

Pollutant Concentration Analysis

Average pollutant levels were compared to identify dominant pollutants.



Findings

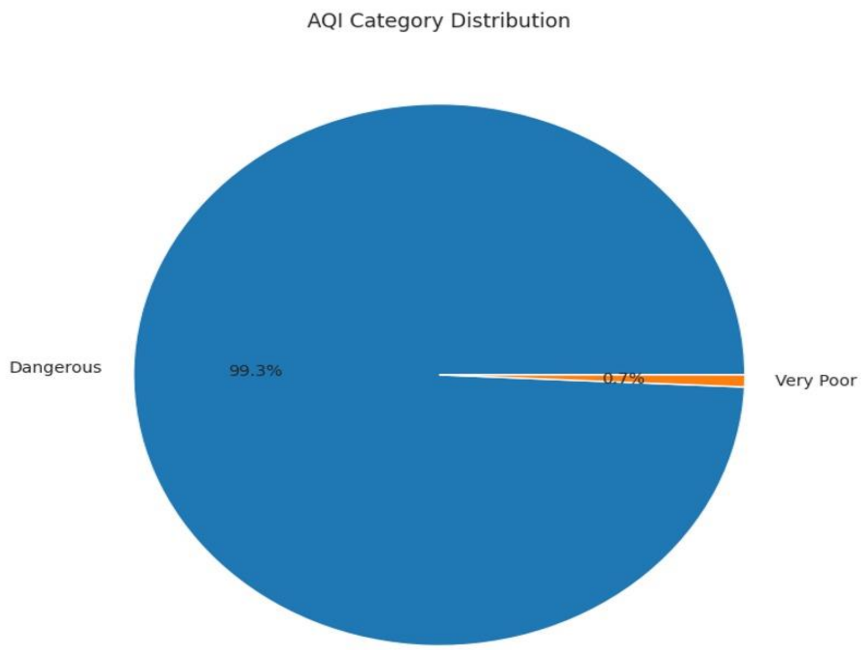
- Carbon Monoxide exhibited the highest concentration among all pollutants.

- PM10 and PM2.5 showed consistently elevated levels.

- Ozone remained comparatively lower than other pollutants.

AQI Classification Analysis

AQI values were grouped into predefined air quality categories.



Findings

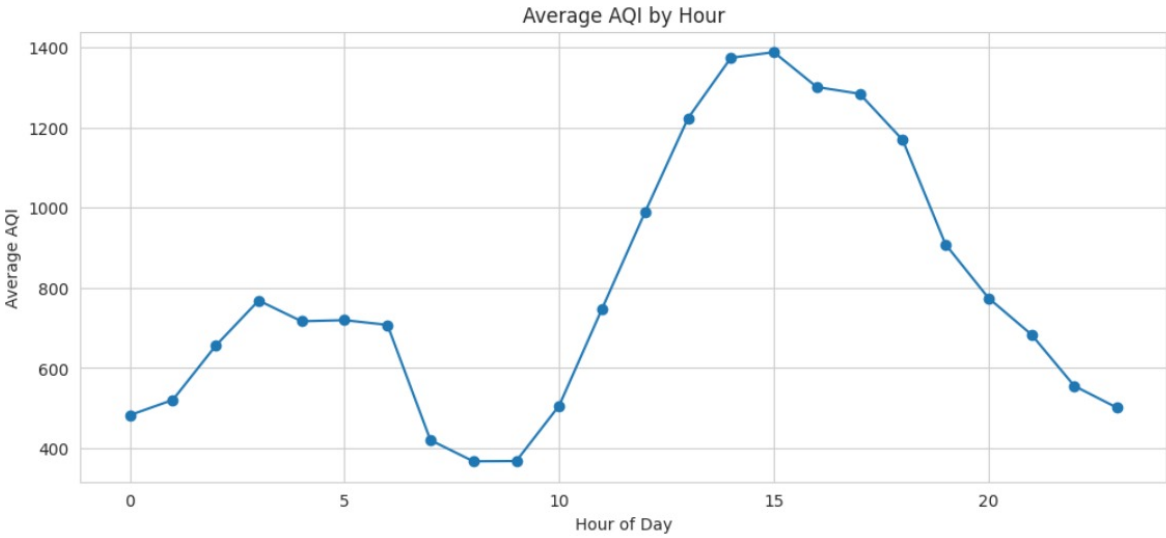
- Most observations belonged to severe pollution categories.

- Healthy air quality conditions were rarely observed.

- Air quality remained poor during most of the study period.

Hourly Pollution Pattern Analysis

Hourly AQI averages were examined to identify peak pollution hour



- Pollution levels varied throughout the day.

- Afternoon hours recorded the highest AQI values.

- Morning periods generally showed comparatively lower pollution levels.

High Pollution Events

The highest AQI observations were investigated.

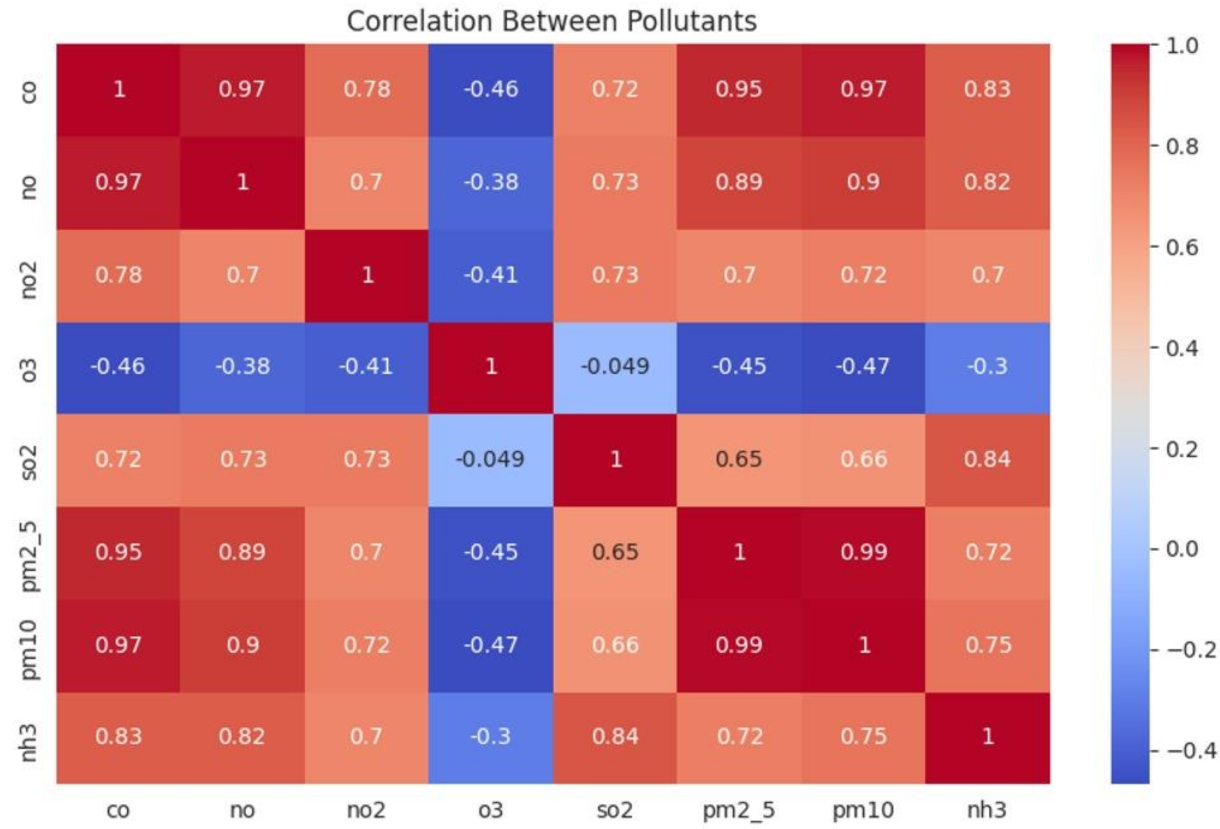
	date	AQI
449	2023-01-19 17:00:00	3372.581667
305	2023-01-13 17:00:00	3327.930000
448	2023-01-19 16:00:00	3308.568333
306	2023-01-13 18:00:00	3299.521667
447	2023-01-19 15:00:00	3272.543333
307	2023-01-13 19:00:00	3185.583333
304	2023-01-13 16:00:00	3182.850000
308	2023-01-13 20:00:00	3062.546667
446	2023-01-19 14:00:00	2987.196667
450	2023-01-19 18:00:00	2940.668333

Findings

- January 19 recorded the most severe pollution event.
- Multiple extreme AQI observations occurred during mid-January.
- Certain periods consistently exhibited elevated pollution levels.

Correlation Analysis

A correlation matrix was generated to study relationships among pollutants.



Findings

- PM2.5 and PM10 displayed a very strong positive relationship.

- Carbon Monoxide showed strong associations with particulate pollutants.

- Ozone demonstrated weaker and sometimes negative relationships with other pollutants.

Major Insights

- Carbon Monoxide emerged as the dominant pollutant in the dataset.

- PM2.5 and PM10 exhibited highly similar behavior patterns.

- Air quality deteriorated significantly during certain periods of January.

- Afternoon hours experienced the highest pollution levels.

- Strong pollutant correlations suggest common sources of emissions.

Conclusion

The analysis provided valuable insights into air pollution patterns in Delhi. The results indicate consistently high pollution levels, with Carbon Monoxide, PM2.5, and PM10 playing major roles in air quality degradation. Hourly trend analysis revealed that pollution levels tend to increase during afternoon periods, while correlation analysis highlighted strong relationships among major pollutants. These findings emphasize the importance of effective pollution-control measures and continuous environmental monitoring to improve public health and urban sustainability.